Chapter 11. Coloring and Matching

Finding a Good Match

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11.1 Introduction

Motivation. Assigning frequencies so nearby towers differ is a vertex colouring.

Four frequencies, ten towers; towers within 50 miles must differ. Join such towers by an edge and colour vertices by frequency.

Proposition 11.1

The four frequencies can be assigned to the ten towers iff the vertices of the graph can be coloured with four colours so that no two adjacent vertices have the same colour.

Proof. An assignment is a 4-colouring, and "towers within 50 miles differ" is "adjacent vertices differ". ■

Definition 11.2 (Chromatic number)

The **chromatic number** $\chi(H)$ is the least k such that the vertices of H can be coloured with k colours with adjacent vertices coloured differently (a **proper colouring**). H is **k-colourable** if $\chi(H) \leq k$.

Example 11.3

$\chi(C_5) = 3$: two colours cannot alternate around an odd cycle; red, blue, red, blue, green works.

We may assume graphs connected and simple (components colour independently; multiple edges add no constraints).

11.2 Bipartite graphs

Motivation. Two colours suffice iff there is no odd cycle; past $n^2/4$ edges, triangles appear in bulk.

Definition 11.4 (Bipartite)

A 2-colourable graph is **bipartite**: its vertex set splits into **colour classes** A, B such that every edge has one end in A and one in B.

Trees (colour by distance parity) and even cycles are bipartite; odd cycles are not.

Theorem 11.5 (König)

A graph is bipartite iff it contains no cycle of odd length.

Proof.

Step 1 (only if). Colours alternate along $A_1 \cdots A_{2m+1}$, so the adjacent A_1, A_{2m+1} agree.

Step 2 (colouring by distance). Let G (wlog connected) have no odd cycle. Fix V; colour W red if $d(V, W)$ is even, blue if odd. An edge PQ inside one colour plus shortest paths $V \to P$, $V \to Q$ gives a closed walk of odd length.

Step 3 (odd closed walks contain odd cycles) [Supplement]. Induction on length: without a repeated vertex it is a cycle; otherwise split at a repeated vertex into two shorter closed walks with odd total length, one of them odd. This contradicts the assumption. ■

Theorem 11.6

A simple bipartite graph on n vertices has at most $\frac{n^2}{4}$ edges if n is even and at most $\frac{n^2-1}{4}$ if n is odd.

Proof. An extremal G with classes of sizes $a, n-a$ has all $a(n-a)$ cross edges, maximised at $a = \lfloor n/2 \rfloor$. ■

The extremal graphs are the **complete bipartite graphs** $K_{a,b}$.

Lemma 11.7 (Mantel)

A simple graph H on $2m$ vertices ($m \geq 2$) with at least $m^2 + 1$ edges contains a triangle.

Proof.

Step 1 ($m = 2$). $H \subseteq K_4$ with ≥ 5 edges is not bipartite (Theorem 11.6), so it has an odd cycle, on 4 vertices a triangle.

Step 2 (induction). Take an edge FG. If $\deg F + \deg G > 2m$, then F, G have at least $2m-1$ neighbours among the other $2m-2$ vertices (besides each other), hence a common neighbour T: triangle FGT. Otherwise deleting F, G removes at most $2m-1$ edges, leaving $2m-2$ vertices and $\geq (m-1)^2 + 1$ edges; apply induction. ■

Theorem 11.8

A simple graph H on $2m$ vertices ($m \geq 2$) with at least $m^2 + 1$ edges contains at least m triangles.

Proof. Assume exactly $m^2 + 1$ edges; induct on m ($m = 2$: K_4 minus an edge has 2 triangles). Lemma 11.7 gives a triangle ABC. Let $2m - 3 + x$ edges join ABC to the other $2m-3$ vertices; by pigeon-hole at least x outside vertices send two edges to ABC, giving $\geq x$ triangles. Let R be the graph on the outside vertices.

Step 1 ($x \geq m-1$). Done.

Step 2 ($1 \leq x < m-1$). At most $3m-5$ edges meet ABC from outside, so R has $\geq m^2 + 1 - (3m-5) - 3 = (m-1)(m-2) + 1$ edges on $2m-3$ vertices. Deleting a minimum-degree vertex leaves R' on $2m-4$ vertices with more than $(m-1)(m-2)\frac{2m-4}{2m-3} = (m-2)^2\frac{2m-2}{2m-3} > (m-2)^2$ edges. Induction gives $m-2$ triangles, plus $x \geq 1$.

Step 3 ($x \leq 0$). At most $2m-3$ edges join ABC to R, so R has $\geq (m-1)^2$ edges. With no such edge, R has $m^2 - 2 \geq (m-1)^2 + 1$ edges and R plus any vertex of ABC has $2m-2$ vertices. Otherwise add to R a vertex of ABC adjacent to R: $2m-2$ vertices, $\geq (m-1)^2+1$ edges. Induction gives $m-1$ further triangles. ■

11.3 Matchings in bipartite graphs

Motivation. Hall's obviously necessary condition is also sufficient; augmenting paths certify maximality.

Jobs and applicants form the classes, qualifications the edges.

Proposition 11.9

All m openings can be filled simultaneously iff the job–applicant graph has m vertex-disjoint edges.

Proof. A complete hiring is exactly a set of m pairwise vertex-disjoint edges "job–person hired". ■

Definition 11.10 (Matching)

A set S of edges of G, no two sharing a vertex, is a **matching** (independent set of edges). It is **perfect** if it covers every vertex.

Definition 11.11 (Matching X into Y)

For a bipartite $G = (X, Y)$, a matching covering all of X is a **perfect matching of X into Y**; we say X **can be matched into Y**.

11.3.1 Bipartite graphs with perfect matchings

For $T \subseteq X$, $N(T) = N_G(T)$ is the set of neighbours of T.

Proposition 11.12

If X can be matched into Y, then $|T| \le |N(T)|$ for all $T \subseteq X$.

Proof. The matching maps T injectively into $N(T)$. ■

Theorem 11.13 (Hall's theorem)

X can be matched into Y iff $|T| \le |N(T)|$ for all $T \subseteq X$.

Proof (Halmos–Vaughan). "Only if" is Proposition 11.12. "If": induct on $|X|$; $|X| \le 1$ is trivial.

Step 1 (strict surplus). If $|T| < |N(T)|$ for all nonempty proper $T \subset X$, take an edge xy and $G' = G - x - y$: $|N_{G'}(A)| \ge |N_G(A)| - 1 \ge |A|$ for $\emptyset \ne A \subseteq X - x$. Match $X - x$ into $Y - y$ by induction; add xy.

Step 2 (a tight set). Otherwise $|B| = |N(B)|$ for some nonempty proper $B \subset X$. Let $G_1 = G[B \cup N(B)]$ and $G_2 = G - (B \cup N(B))$. Hall's condition holds in G_1 ($N_{G_1}(T) = N_G(T)$), and in G_2, since $N_G(U \cup B) = N_{G_2}(U) \sqcup N_G(B)$ for $U \subseteq X - B$,

$$|N_{G_2}(U)| = |N_G(U \cup B)| - |N_G(B)| \ge |U| + |B| - |B| = |U|.$$

Induction matches B onto $N(B)$ and $X - B$ into $Y - N(B)$. ■

A matching is **maximal** if no edge can be added, **maximum** if none is larger. Maximum implies maximal, not conversely (the middle edge of a 3-edge path).

Definition 11.14 (Alternating path)

For a matching M, a path $v_1 v_2 \cdots v_r$ is **M-alternating** if its edges alternate between M and not M.

Definition 11.15 (Augmenting path)

An M-alternating path is **M-augmenting** if it starts and ends at vertices not covered by M.

$M \mapsto M \triangle P$ for an augmenting path P adds one edge.



Figure. *An M-augmenting path (bold: M); swapping bold and dashed edges enlarges the matching.*

Theorem 11.16 (Berge)

A matching M in a simple graph is maximum iff there is no M-augmenting path.

Proof. "Only if": an augmenting path enlarges M. "If": for a maximum M', the components of $M \triangle M'$ are M/M'-alternating paths and even cycles. A path with more M'-edges would be M-augmenting, one with more M-edges M'-augmenting. So each component is balanced and $|M| = |M'|$. ∎

11.3.2 Stable matchings in bipartite graphs

n candidates rank n jobs, and each hiring manager ranks all candidates.

Definition 11.17 (Stable matching)

In a complete bipartite $G(X, Y)$ where every vertex ranks the other side, a perfect matching M is **stable** if there are no $x \in X$, $y' \in Y$ such that (a) $xy \in M$ but x prefers y' to y, and (b) $x'y' \in M$ but y' prefers x to x' (no pair would both rather be together).

Gale–Shapley (deferred acceptance). Each unmatched candidate applies to the best job not yet tried; each manager tentatively keeps the best applicant so far and rejects the rest. It ends with everyone placed (an unplaced candidate always has an untried job).

The result is stable. Let $xy, x'y' \in M$, x prefer y', y' prefer x to x'. Then x applied to y' earlier, and y' either accepted and later dropped x or rejected x outright, each time for someone better; its final x' is at least as good, a contradiction. Unequal sides: add fictitious members ranked last. In non-bipartite graphs stable matchings need not exist.

Remark

Remark[Algorithmic note] $O(n^2)$ applications: each candidate applies to each job at most once.

11.4 More than two colours

Motivation. Turán: more edges than the balanced complete k-partite graph force K_{k+1}.

Write $n = tk + r$, $0 \le r \le k-1$; split n vertices into r classes of size $t+1$ and $k-r$ of size t, joining vertices in different classes. This **complete k-partite (Turán) graph** H is k-colourable with

$$T(n,k) = \frac{k-1}{2k}\, n^2 - \frac{r(k-r)}{2k} \tag{11.1}$$

edges. [Supplement] *Reason:* $\frac{1}{2}(n^2 - r(t+1)^2 - (k-r)t^2)$ with $n = tk + r$ gives (11.1).

Theorem 11.18 (Turán)

A simple graph on n vertices with more than $T(n,k)$ edges contains K_{k+1}; in particular it is not k-colourable.

Proof.

Step 1. Let G be K_{k+1}-free on n vertices with the most edges; show $e(G) \le T(n,k)$ by induction on $t = \lfloor n/k \rfloor$. For $t = 0$, $T(n,k) = \binom{n}{2}$.

Step 2. For $t \ge 1$, G contains a K_k, S: if G is complete, $n \ge k$; otherwise adding a missing edge creates a K_{k+1}, and its other $k-1$ vertices plus one endpoint form a K_k in G.

Step 3. S has $\binom{k}{2}$ edges, each outside vertex has $\le k-1$ neighbours in S, and $G-S$ on $(t-1)k+r$ vertices has $\le T(n-k,k)$ edges by induction:

$$e(G) \le \binom{k}{2} + (n-k)(k-1) + T(n-k,k) = T(n,k). \tag{11.2}$$

Step 4 (the identity (11.2)) [Supplement]. Removing one vertex from each class of H leaves the Turán graph on $n-k$ vertices; the removed edges are the $\binom{k}{2}$ among removed vertices and $k-1$ per remaining vertex. ■

Odd cycles of length ≥ 5 have $\chi = 3$ without K_3; odd cycles and complete graphs are the only obstacles:

Theorem 11.19 (Brooks)

If G is connected, neither a complete graph nor an odd cycle, and all degrees are at most d with $d \ge 3$, then $\chi(G) \le d$.

Proof [Supplement] (Lovász).

Step 1 (greedy towards a root). If $\deg w < d$, colour greedily from $[d]$ in non-increasing distance from w. Each $v \ne w$ has a later-coloured neighbour, so $\le d-1$ coloured neighbours; w has $< d$ neighbours. So assume G is d-regular.

Step 2 (a cut vertex). If v is a cut vertex with components C_i of $G-v$, each $G_i = G[C_i \cup \{v\}]$ has $\deg v < d$ and is d-colourable by Step 1 (or complete on $\le d$ vertices). Permute colours to agree at v and combine.

Step 3 (the key triple). Let G be 2-connected; find w with non-adjacent neighbours u, v and $G-u-v$ connected.

If G is 3-connected: G is not complete, so some w on a shortest path between non-adjacent vertices has non-adjacent neighbours u, v, and $G-u-v$ is connected.

If $\{w,y\}$ is a 2-vertex cut: $G-w$ has cut vertex y, hence ≥ 2 leaf blocks, each with a non-cut vertex adjacent to w (else G has a cut vertex). Take such u, v in different leaf blocks (non-adjacent). Degrees in $G-w$ are $\ge d-1 \ge 2$, so leaf blocks are not single edges and $G-w-u-v$ is connected; w keeps a neighbour there ($d \ge 3$), so $G-u-v$ is connected.

Step 4 (colouring). Colour u, v with 1; colour the rest greedily from $[d]$ in non-increasing distance from w in $G-u-v$, ending with w. Each $v' \ne w$ has $\le d-1$ coloured neighbours; the d neighbours of w use $\le d-1$ colours since u, v agree. ■

For every n there are triangle-free graphs with $\chi = n$.

11.5 Matchings in graphs that are not bipartite

Motivation. Tutte: deleting s vertices leaves at most s odd components; sufficiency via saturated graphs.

$c_o(G-S)$ is the number of odd components of $G-S$.

Theorem 11.20 (Tutte)

G has a perfect matching iff $c_o(G-S) \le |S|$ for every vertex set S.

Only if. Each odd component of $G-S$ has a vertex matched into S, to distinct partners.

G is **saturated non-factorizable** if it has no perfect matching but adding any edge creates one.

Lemma 11.21

If G is saturated non-factorizable and S is the set of vertices adjacent to all other vertices, then the components of $G-S$ are complete graphs.

Proof. Let $ab, bc \in E(G-S)$, $ac \notin E$, and $d \not\sim b$ ($b \notin S$). Perfect matchings $F_1 \ni ac$ of $G+ac$ and $F_2 \ni bd$ of $G+bd$ give $F_1 \triangle F_2$, a union of even alternating cycles; let $C_1 \ni ac$, $C_2 \ni bd$.

Step (a) $C_1 \neq C_2$. $F_1 \triangle C_1$ is a perfect matching of G, a contradiction.

Step (b) $C_1 = C_2$. Traverse C_1 from b through d to the first of a, c, say a, along P. Then $F_2 \triangle (P \cup ab)$ is a perfect matching of G avoiding bd, a contradiction. ■

Theorem 11.22

G is saturated non-factorizable iff either (a) G is complete with an odd number of vertices, or (b) G has an even number of vertices and consists of vertex-disjoint complete graphs $S_0, G_1, \dots, G_k$ with $k = |S_0| + 2$, each $|G_i|$ odd, and every vertex of every G_i joined to every vertex of S_0.

Proof.

Step 1 (odd order). Odd graphs have no perfect matching, and only K_n admits no new edge.

Step 2 (even order: structure). With $S_0 = S$ as in Lemma 11.21, S_0 and the G_i are complete and S_0 is joined to all. If $c_o(G-S) \le |S|$: match one vertex of each odd G_i into S, the rest of each G_i internally, the (even) rest of S among itself, a contradiction. So $c_o \ge |S|+2$ by parity. If $c_o > |S|+2$ or an even component existed, adding an edge between two components keeps $c_o(G'-S) > |S|$, contradicting saturation. So $k = |S_0| + 2$, all odd.

Step 3 (conversely) [Supplement]. In (b), $c_o(G-S_0) = |S_0| + 2 > |S_0|$: no perfect matching. A missing edge $g_i g_j$ joins two G_i; using it, the other $|S_0|$ odd components each send one vertex into S_0 and the rest match inside: a perfect matching. ■

Proof of Theorem 11.20 ("if", Hetyei–Lovász). Let G satisfy Tutte's condition without a perfect matching; $S = \emptyset$ gives $|V(G)|$ even. Saturate to $G' \supseteq G$ as in Theorem 11.22(b), with universal set S' and $|S'|+2$ odd complete components of $G'-S'$. Each contains an odd component of $G-S'$, so $c_o(G-S') \ge |S'|+2 > |S'|$, a contradiction. ■

11.6 Selected exercises

Motivation. Hall's condition via a single double count.

Exercise 11.4 (A degree condition for Hall)

Let G be a bipartite graph with colour classes X, Y in which every vertex of X has degree at least as large as every vertex of Y, and every vertex of X has at least one neighbour. Prove that X can be matched into Y.

Solution. Let $a = \min_{x\in X} \deg x \geq b = \max_{y\in Y} \deg y$, $a \geq 1$.

Step 1 (double count). For $T \subseteq X$, the edges E_T at T satisfy $a|T| \leq |E_T| \leq b\,|N(T)|$.

Step 2 (Hall's condition). So $a|T| \leq a|N(T)|$, $|T| \leq |N(T)|$; apply Theorem 11.13.

$a \geq 1$ only excludes the edgeless case $a = b = 0$. ■

Exercise 11.9 (Magic squares are sums of permutation matrices)

Let A be an $n \times n$ matrix with nonnegative integer entries in which every row and every column has the same sum r. Prove that A is a sum of r permutation matrices.

Solution. Induction on r; $r = 0$ is the empty sum. Let $r \geq 1$.

Step 1 (a bipartite graph). Rows $X = \{x_i\}$, columns $Y = \{y_j\}$, $x_iy_j \in E$ iff $a_{ij} > 0$.

Step 2 (Hall's condition). For $T \subseteq X$, the rows of T sum to $r|T|$, all their nonzero entries lie in columns of $N(T)$, which sum to $r|N(T)|$; so $|T| \leq |N(T)|$.

Step 3 (one permutation matrix). Theorem 11.13 gives a perfect matching $x_iy_{\sigma(i)}$, so $a_{i\sigma(i)} \geq 1$; let P_σ be its permutation matrix.

Step 4 (induction). $A - P_\sigma \geq 0$ has line sums $r-1$, so it is a sum of $r-1$ permutation matrices. ■